

If you instantiate the `JCheckBox` without an image, it will get the familiar square box. If you instantiate the `JCheckBox` with an image, then the square box will be omitted and no default square box will be created. When you create a menu, you may also add a menu item that is a check box. For this purpose, you would use a:

```
JCheckBoxMenuItem
```

This control follows the standard L & F as other objects. For example, you can set the Mnemonic as follows:

```
JCheckBox ans = new JCheckBox( "Yes" );  
ans.setMnemonic( KeyEvent.VK_Y );
```

This is how to change the status of the control, and of course, this would trigger an `itemStateChanged` event.

```
ans.setSelected( true );
```

10.8 JRadioButton

These are two-state buttons that reflect their status by having a dot in a circular hole. The interior of the hole is either white or gray, depending on the L&F. When organized in a structure called a `ButtonGroup`, only one `ButtonGroup` in the group can be selected at one time. `JRadioButtons` are Mutually Exclusive.

Just as with the `JCheckBoxes`, `JRadioButtons` have many Constructors, taking an `Icon`, a `String` caption, and a preset boolean for either `Selected` or `Deselected`. When a `JRadioButton` is `Selected`, it fires an `ItemEvent`. Therefore, if you use a `JRadioButton`, you must implement the `ItemListener` interface as with the `JCheckBox`. The logical relationship between the buttons gathered together to form a button group is maintained by the object `ButtonGroup`.

Warning: The `ButtonGroup` is NOT a GUI object. It cannot appear on the GUI and so you do NOT ever add a `ButtonGroup` object to the `Container`. To do so is a syntax error. Joining your `JRadioButtons` into a `ButtonGroup` is entirely separate from adding the `JRadioButtons` to the GUI's `Container`.

```
Container c = getContentPane();  
c.setLayout( new FlowLayout );  
JRadioButton ital = new JRadioButton( "Italic" );  
JRadioButton bold = new JRadioButton( "Bold" );  
JRadioButton undr = new JRadioButton( "Underline" );  
c.add( ital );  
c.add( bold );
```